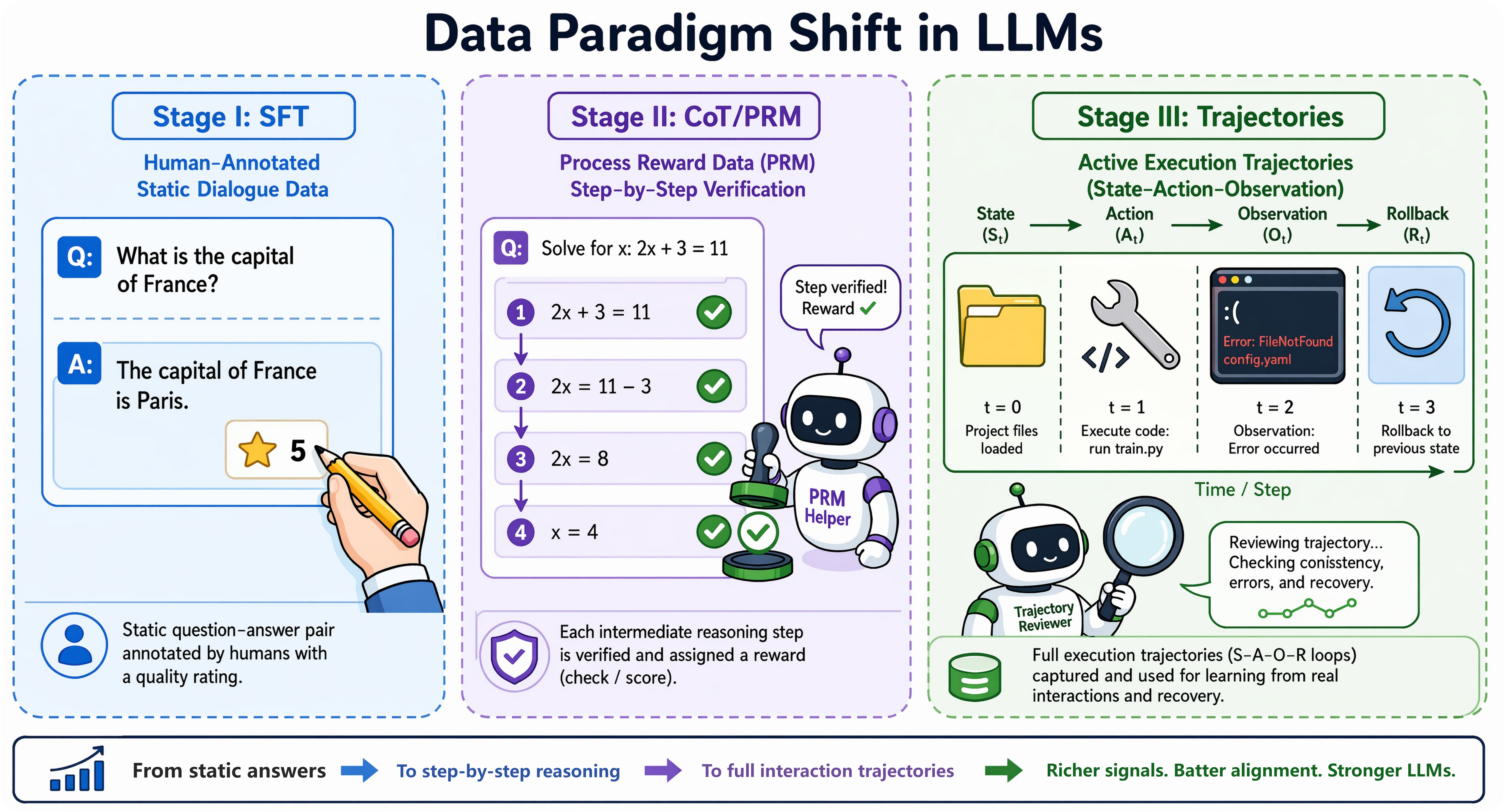
Data Paradigm Shift in LLMs
Stage I: SFT
Human-Annotated
Static Dialogue Data
Q: What is the capital of France?
A: The capital of France is Paris.
5
Static question-answer pair annotated by humans with a quality rating.
Stage II: CoT/PRM
Process Reward Data (PRM)
Step-by-Step Verification
Q: Solve for x: 2x + 3 = 11
1 2x + 3 = 11
2 2x = 11 − 3
3 2x = 8
4 x = 4
Step verified! Reward
PRM Helper
Each intermediate reasoning step is verified and assigned a reward (check / score).
Stage III: Trajectories
Active Execution Trajectories
(State-Action-Observation)
State (S_t)
Action (A_t)
Observation (O_t)
Rollback (R_t)
:(
Error: FileNotFound
config.yaml
t = 0
Project files loaded
t = 1
Execute code: run train.py
t = 2
Observation: Error occurred
t = 3
Rollback to previous state
Time / Step
Trajectory Reviewer
Reviewing trajectory... Checking consisstency, errors, and recovery.
Full execution trajectories (S-A-O-R loops) captured and used for learning from real interactions and recovery.
From static answers
To step-by-step reasoning
To full interaction trajectories
Richer signals. Batter alignment. Stronger LLMs.